

# Large Adjustment Generator

## User Guide

Payroll Operations Team  
Version 2.1 | May 2026

LARGE ADJ GENERATOR

v3.0

Clear Data

PAYROLL OPS SUITE

Deploy

Open User Guide (PDF)

SettingsEmployee DataData DumpAdditional TaxesCustom DataResultsDashboard

SETTINGS

Tax Year

2026

Mode

Net Pay

Gross Up

Employer Taxes

Yes

Supplemental Rates

No

Split CSV

No

Only Income Taxes

No

FUTA

Yes

SS wage base:

\$184,500

Applies supplemental FIT + state rates

Caps each CSV under \$10K employer debit

Skips non -450 state taxes (CA keeps all)

Excludes 00-402 when No

Adjustment Date

05/15/2026

CID

e.g. C001

Notes

Add any notes here...

PER-EMPLOYEE TAXES

Taxes are pre-loaded from the employee's state. Add or adjust as needed per employee.

Paste employee data on the Employee Data tab first to configure per-employee taxes.

Page 1/13

# 1. Overview

---

The Large Adjustment Generator (LAG) is an enterprise payroll tax calculation and reporting tool built for the Payroll Operations team. It computes federal, state, and local tax adjustments for one or more employees, supports gross-up calculations, and generates compliance-ready CSV exports for upload to the payroll system.

## Key Capabilities

- Calculate federal taxes: Social Security (6.2%), Medicare (1.45%), Additional Medicare (0.9%), FUTA (0.6%)
- Calculate state withholding and employer unemployment (SUI) for all 50 states + DC, PR, VI
- Supplemental state withholding rates (year-aware, auto-applied to -450 codes)
- Gross-up mode: solve for the gross pay that yields a specific net amount
- YTD wage base tracking to correctly cap Social Security and SUI contributions
- Custom tax reconciliation with 70/30 split (70% federal, 30% state)
- Split CSV mode: partition exports so each file stays under \$10,000 employer debit
- Interactive dashboard with geographic heatmap, tax composition charts, and drill-down
- Client-facing PDF report generation with plain-English tax explanations
- Batch processing for multiple employees across multiple states simultaneously

## Supported Modes

**Net Pay Mode:** You enter the gross pay amount. The tool calculates all taxes and reports the resulting net pay.

**Gross Up Mode:** You enter the desired net pay amount. The tool uses binary search to find the gross pay that, after all taxes are deducted, yields exactly that net amount.

# 2. Getting Started

---

## Accessing the Tool

The Large Adjustment Generator runs on port 8505. Navigate to <http://localhost:8505> in your browser.

## Authentication

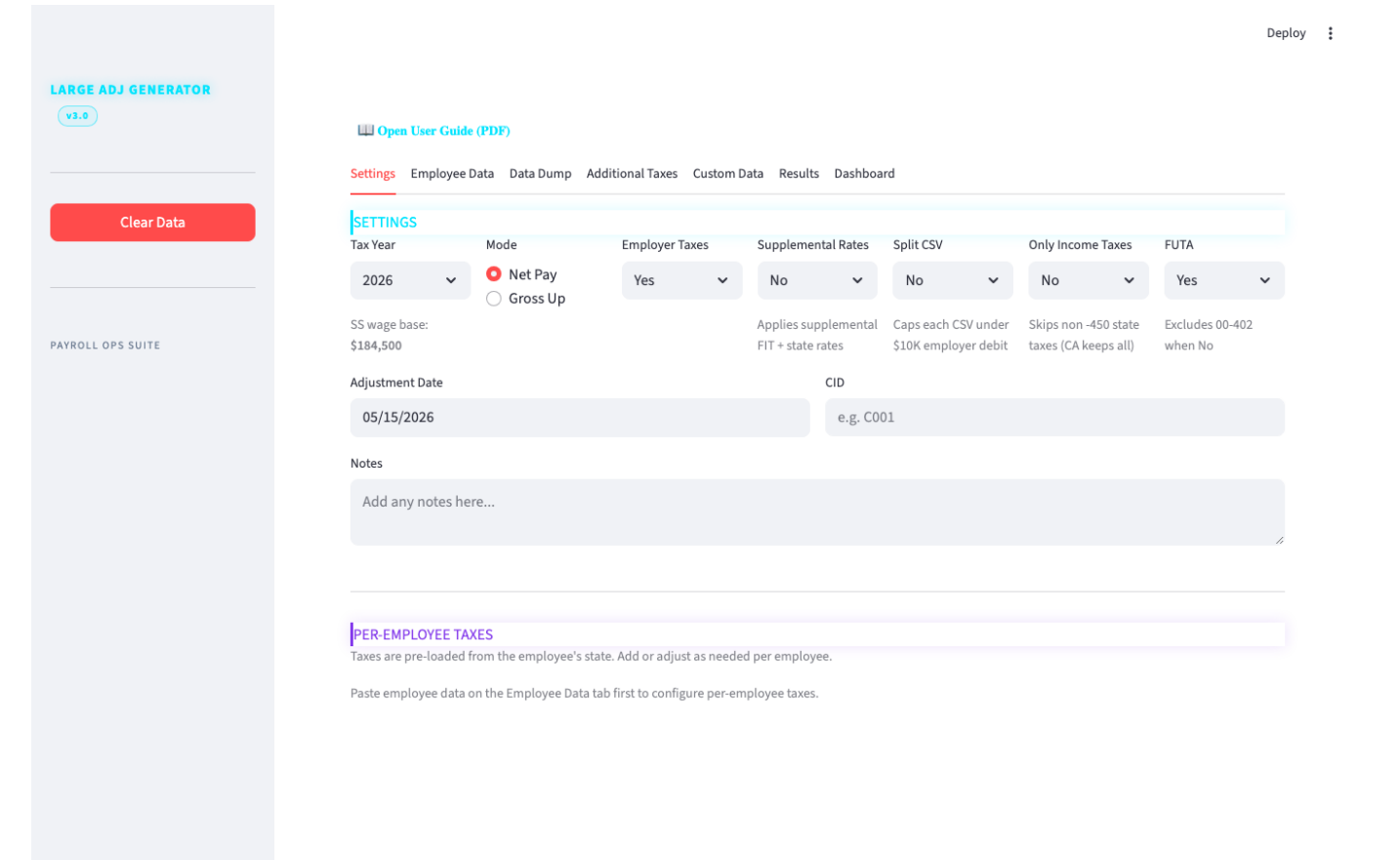
Enter the team password when prompted. The password is shared among the Payroll Operations team. After authentication, you will see the main interface with 7 tabs.

## Recommended Workflow

1. Configure settings (tax year, mode, employer taxes, toggles)
2. Upload YTD Medicare wages via Data Dump tab (optional but recommended)
3. Add any additional/custom taxes via Additional Taxes tab (optional)
4. Paste employee data (MID, Amount, State) in Employee Data tab
5. Upload actual tax withheld via Custom Data tab (optional, for reconciliation)
6. Review/adjust per-employee tax rates in Settings tab

- 7. Generate results in the Results tab
- 8. Download CSV export from Results tab
- 9. Review dashboard analytics and generate client PDF (Dashboard tab)

### 3. Settings Tab



Settings tab - main configuration panel

#### Core Settings

##### Tax Year

Select 2023-2026. Determines Social Security wage base: 2023=\$160,200 | 2024=\$168,600 | 2025=\$176,100 | 2026=\$184,500

##### Mode

"Net Pay" (input is gross) or "Gross Up" (input is desired net pay)

##### Employer Taxes

Yes/No. When Yes, includes employer SS, Medicare, FUTA, and SUI in calculations and CSV

##### Supplemental Rates

When Yes, auto-applies state-specific supplemental withholding rates to -450 tax codes

##### Split CSV

When Yes, partitions CSV exports so each file's employer debit stays under \$10,000

##### Only Income Taxes

When Yes, filters employee taxes to state withholding codes (-450) only. California always included regardless

##### FUTA

Yes/No. Controls whether FUTA (code 00-402) appears in calculations and CSV export

Adjustment Date

Date stamp for the CSV. Defaults to next Friday

CID

Company identifier included in all CSV rows

Notes

Free-form text included in CSV reference field

Per-Employee Tax Configuration

After entering employee data, each employee gets an expandable section in the Settings tab. This section shows:

Employee Withholdings: State-specific taxes auto-populated based on the employee's state. Supplemental rates are auto-applied when the toggle is on. You can adjust rates, add new tax rows, or remove rows.

Employer Contributions: Corresponding employer taxes (SUI, etc.) with wage base limits auto-calculated from the SUI\_WAGE\_BASES table for the selected year.

Tax rows auto-refresh when you change: state, year, employer tax toggle, supplemental rates toggle, income-only filter, or additional taxes.

4. Employee Data Tab

LARGE ADJ GENERATOR

v3.0

Clear Data

PAYROLL OPS SUITE

Deploy

Open User Guide (PDF)

SettingsEmployee DataData DumpAdditional TaxesCustom DataResultsDashboard

EMPLOYEE DATA

Paste a table with columns: MID, Amount, State. YTD Medicare wages are pulled from the Data Dump tab.

| MID    | Amount  | State      |
|--------|---------|------------|
| M12345 | 5000.00 | New York   |
| M67890 | 3200.00 | California |

Employee Data tab - paste employee information

Input Format

Paste a tab-separated or comma-separated table with 3 columns:

MID | Amount | State

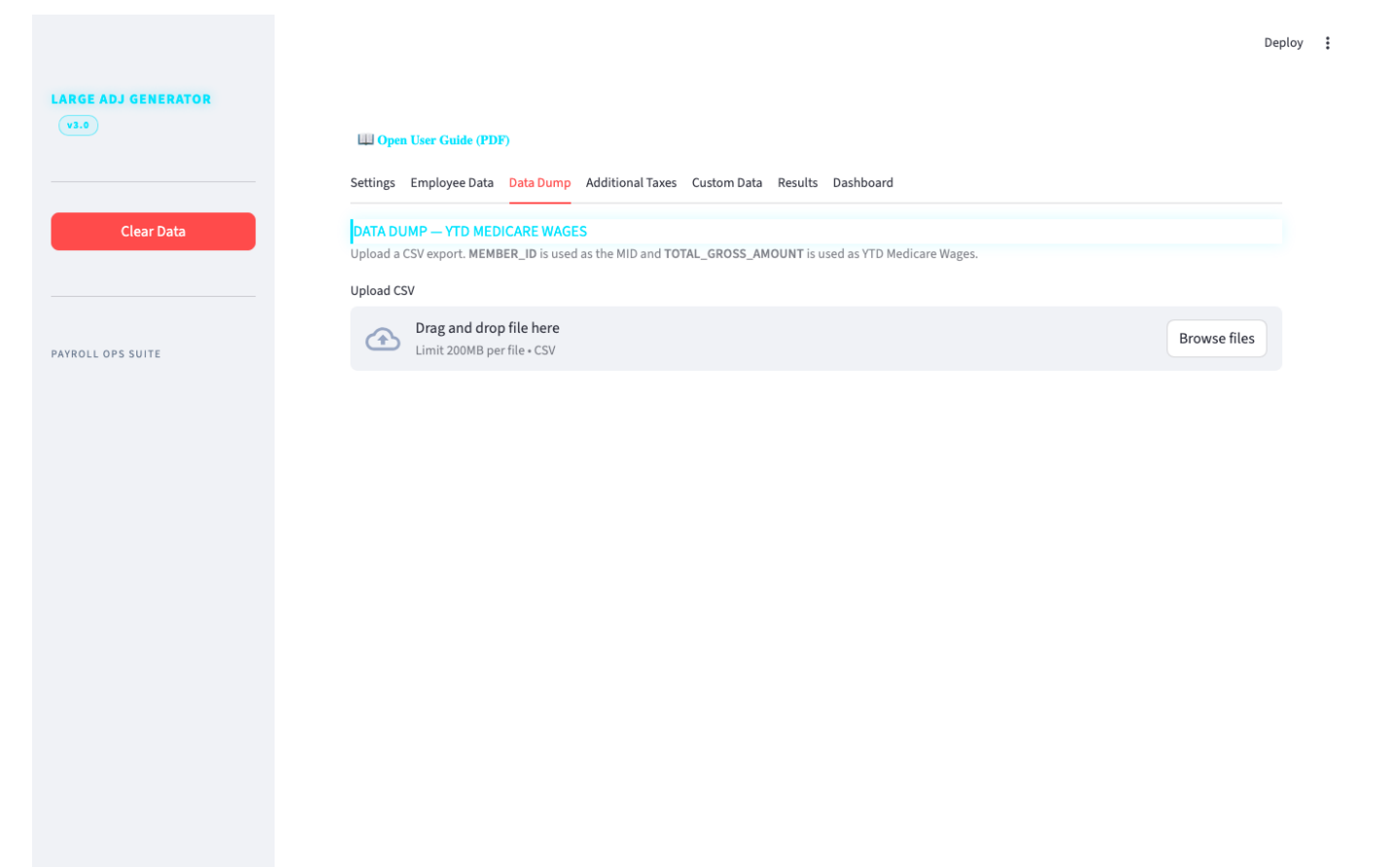
Example:

M12345 5000.00 California  
M67890 3200.00 NY  
M11111 8500.00 TX

Automatic Processing

- State abbreviations automatically expanded (NY -> New York, CA -> California, etc.)
- Duplicate MIDs are aggregated: amounts are summed, first-seen state is retained
- Currency formatting (\$, commas) handled automatically
- Tab or comma separators both accepted

5. Data Dump Tab (YTD Medicare Wages)



Data Dump tab - upload YTD wage data

Upload a CSV file containing year-to-date Medicare wages for employees. This data is critical for correctly capping Social Security contributions when an employee is near or above the annual wage base.

Required CSV Format

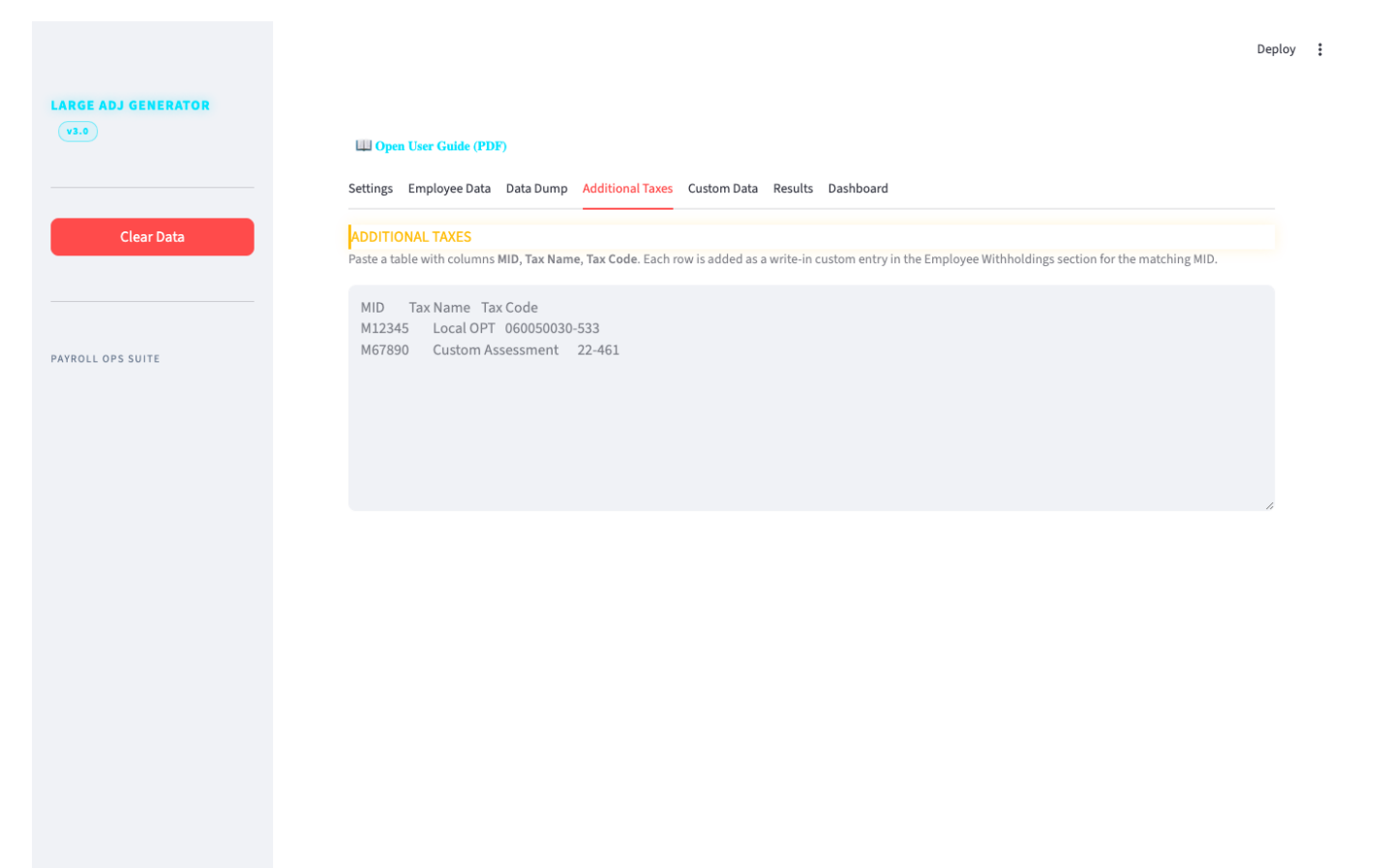
- The CSV must contain these columns:
- MEMBER\_ID: Employee identifier (matched to MID)
  - TOTAL\_GROSS\_AMOUNT: Year-to-date gross wages

Additional columns are ignored. Duplicate MIDs are summed. Currency formatting (\$, commas) is handled automatically.

Why This Matters

Without YTD data, the tool assumes \$0 year-to-date wages. This means Social Security tax will be calculated on the full adjustment amount even if the employee has already exceeded the annual wage base (\$184,500 for 2026). Always upload YTD data for accurate results.

6. Additional Taxes Tab



Additional Taxes tab - custom per-employee taxes

Add custom or non-standard taxes that are not in the built-in state/federal tax library. These are injected into the per-employee tax configuration in the Settings tab.

Input Format

Paste a table with 3 columns:  
MID | Tax Name | Tax Code

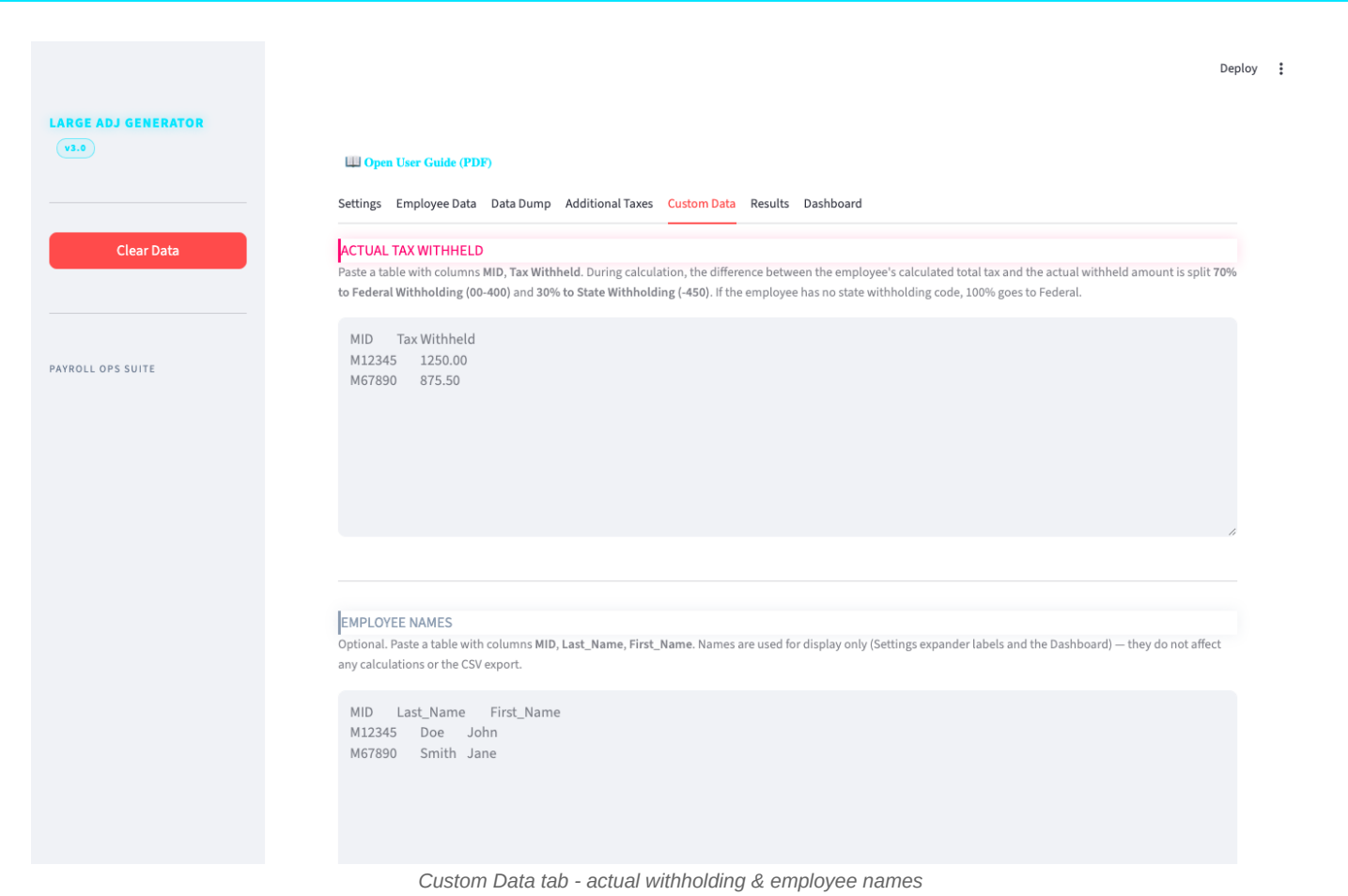
Example:  
M12345 Aurora OPT 060050030-533  
M67890 Denver OPT 060310140-533

Behavior

- Taxes appear as additional rows in the Settings tab per-employee section

- Duplicate (MID, tax code) pairs are skipped to prevent double-counting
- Rate defaults to 0% - you must set the rate in the Settings tab
- Common use cases: local occupational privilege taxes, city taxes, special assessments

## 7. Custom Data Tab (Reconciliation)



### Actual Tax Withheld (Reconciliation)

Enter the actual total tax withheld for employees when you need to reconcile calculated vs. actual amounts. The tool computes the variance and applies it using a 70/30 split between federal and state taxes.

Input format (2 columns):

MID | Tax Withheld

70/30 Reconciliation Logic:

1. Calculates  $\text{diff} = \text{actual\_withheld} - \text{calculated\_total}$
2. Splits 70% of the variance to Federal Withholding (00-400)
3. Splits 30% of the variance to State Withholding (-450 code)
4. If no state -450 code exists, 100% goes to Federal (00-400)
5. Creates tax lines as needed if they don't already exist

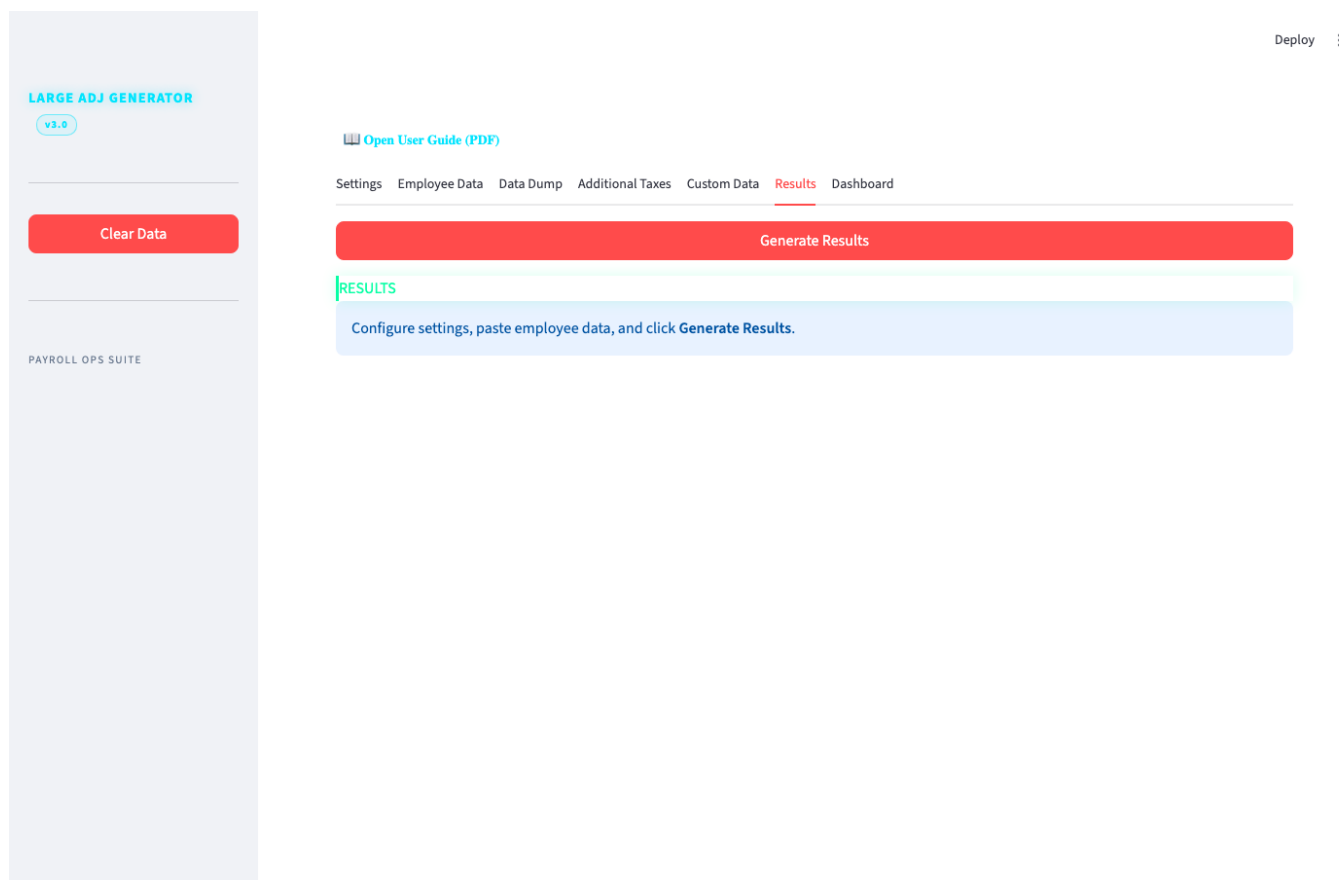
### Employee Names (Optional)

Paste a table with 3 columns:

MID | Last\_Name | First\_Name

Names are displayed in Settings tab expanders, dashboard dropdowns, and client PDF reports. They do not affect calculations or CSV export.

## 8. Results Tab



Results tab - generate and export

### Generating Results

Click the "Generate Results" button to run tax calculations for all employees. The tool validates that employee data is present, then computes all federal, state, and custom taxes based on your settings.

### Results Display

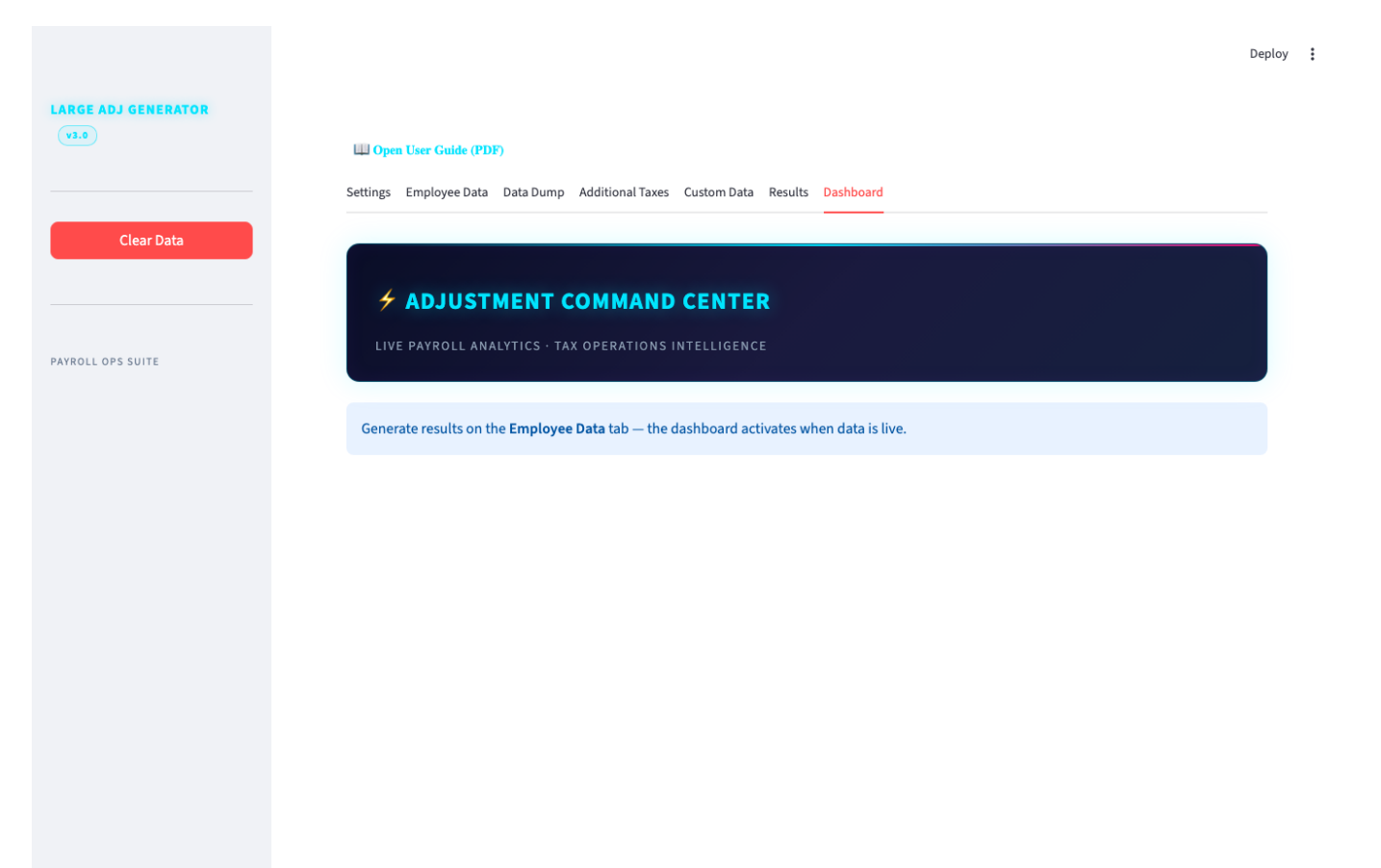
- Summary metrics: employee count, total gross, total net, total employee tax
- Per-employee results table with currency formatting
- Expandable breakdown per employee: SS, Medicare, Additional Medicare, FUTA details

### CSV Export

The CSV export section generates a compliance-ready file for payroll system upload.

- One Clearing/Credit row per tax per employee (employee and employer taxes)
- One consolidated Company/Debit row summing all credits
  - 15 columns: account\_type, entry\_type, adjustment\_date, amount, cid, tax\_code, member\_id, taxable\_amount, subj\_gross, adjusted\_gross, adjusted\_supl\_gross, gross\_earnings\_amount, reference\_type, reference\_id, notes
- BOM included for Excel compatibility
- Split CSV mode: multiple files, each under \$10,000 employer debit

## 9. Dashboard Tab



Dashboard tab - interactive analytics

### KPI Row

- Top-level metrics at a glance:
- Employees: count + number of states
  - Total Gross: total payroll outflow
  - Total Net: net pay + effective withholding rate
  - Employer Cost: SS + Medicare + FUTA + other employer taxes
  - Total Debit: combined employee + employer tax amount

### Employee Deep Dive

- Select an employee from the dropdown (or click a bar in the chart) to see:
- Individual KPI cards (gross, net, EE tax, ER cost, total debit)
  - Tax line items table (all employee + employer taxes with rates and amounts)
  - Withholding adjustment explainer (if Custom Data reconciliation was applied)
  - Tax composition pie/donut charts

### Geographic Heatmap

US map showing employee distribution by state. Bubble size indicates employee count. Click to drill into state-specific tax composition.

### Client Report PDF

Generate a professional client-facing PDF with:

- Summary metrics (no tax codes - plain English)
- Per-employee sections with tax tables
- Withholding adjustment explanations
- Professional formatting with page breaks

## State Tax Export

Download a separate CSV aggregating taxes by state for reconciliation purposes. This export can be fed directly into the State Amendments tool (port 8508).

## 10. Supplemental Rates

---

When the Supplemental Rates toggle is enabled in Settings, the tool automatically applies state-specific supplemental withholding rates to the -450 (state income tax) code for each employee. Rates are year-aware and update when you change the Tax Year.

### How It Works

1. Toggle 'Supplemental Rates' to Yes in the Settings tab.
2. The tool looks up the employee's state and selected tax year.
3. The appropriate supplemental rate is pre-filled for the -450 code.
4. You can still override the rate manually per employee.

### Special State Handling

#### California

Default rate: 6.6%. Note: stock option income uses 10.23% - adjust manually if applicable.

#### Vermont

Rate is 30% of federal supplemental (22%), calculated as 6.6%.

#### Wisconsin

Graduated brackets based on gross amount: 3.5% (0-14,320), 4.4% (14,320-28,640), 5.3% (28,640-315,310), 6.53% (315,310-693,860), 7.65% (693,860+).

#### Maryland

Rate is 0% for 2025 (check manually). Rate is 6.5% for 2026.

### Rate Changes 2025 to 2026

Several states changed their supplemental withholding rates for 2026. Key changes include:

- Georgia: 5.39% -> 5.19%
- Idaho: 5.695% -> 5.3%
- Indiana: 3.0% -> 2.95%
- Kentucky: 4.0% -> 3.5%
- Maryland: 0.0% -> 6.5%
- Mississippi: 4.4% -> 4.0%
- Nebraska: 5.0% -> 3.5%
- North Carolina: 4.35% -> 4.09%
- Ohio: 3.5% -> 2.75%
- Oklahoma: 4.75% -> 4.5%

- South Carolina: 6.2% -> 6.0%
- Utah: 4.55% -> 4.5%

## 11. Calculation Logic

---

### Social Security (6.2%)

Rate: 6.2% (employee) + 6.2% (employer)

Annual wage base (2026): \$184,500

Calculation: taxable = min(gross, max(0, wage\_base - YTD\_SS))

Amount = taxable x 6.2%

If an employee's YTD wages already exceed the wage base, no Social Security tax is calculated on the adjustment.

### Medicare (1.45%)

Rate: 1.45% (employee) + 1.45% (employer)

No wage base cap - applies to all wages.

Amount = gross x 1.45%

### Additional Medicare (0.9%)

Rate: 0.9% (employee only - no employer match)

Threshold: \$200,000 cumulative annual wages

Calculation: Only applies to wages above \$200,000 when combining YTD + current gross.

Amount = max(0, (YTD + gross) - \$200,000) x 0.9% (minus any already-taxed portion)

### FUTA (0.6%)

Rate: 0.6% (employer only)

Wage base: \$7,000

Calculation: taxable = min(gross, max(0, \$7,000 - YTD))

Amount = taxable x 0.6%

Can be toggled off in Settings if not applicable.

### 70/30 Reconciliation Split

When Custom Data provides actual tax withheld amounts, the tool reconciles the variance between calculated and actual using a 70/30 split:

- 70% of the variance is applied to Federal Income Tax (00-400)
- 30% of the variance is applied to State Income Tax (-450 code)

This ensures the adjustment distributes the withholding difference proportionally between federal and state obligations. If no state -450 code exists for the employee, the full 100% goes to federal.

### Gross-Up Algorithm

When Mode = "Gross Up", the tool uses binary search to find the gross pay that produces the desired net:

1. Search range: [desired\_net, desired\_net x 10]
2. For each midpoint, calculate all taxes and check resulting net

- 3. Converge within \$0.001 (up to 300 iterations)
- 4. Result: the gross amount where net = desired amount after all taxes

State & Custom Taxes

State taxes use configurable rates set in the per-employee Settings section. Each tax row specifies: name, code, rate (%), and optional wage base limit.

If a wage base limit is set (e.g., SUI): taxable = min(gross, remaining\_room)  
Otherwise: taxable = gross  
Amount = taxable x rate%

12. Appendix

Federal Tax Codes

|        |                                |
|--------|--------------------------------|
| 00-400 | Federal Income Tax Withholding |
| 00-402 | FUTA (Federal Unemployment)    |
| 00-403 | Social Security - Employee     |
| 00-404 | Social Security - Employer     |
| 00-406 | Medicare - Employee            |
| 00-407 | Medicare - Employer            |
| 00-901 | Additional Medicare            |

State Tax Code Pattern

- XX-450: State income tax withholding (employee)
- XX-459: State unemployment (employer)
- XX-461: Employment training tax
- XX-466: State disability insurance
- XX-468: Paid family leave

Where XX is the state number (01=Alabama through 51=Wyoming, plus territories).

Social Security Wage Bases by Year

|      |           |
|------|-----------|
| 2023 | \$160,200 |
| 2024 | \$168,600 |
| 2025 | \$176,100 |
| 2026 | \$184,500 |

Key SUI Wage Bases (2026)

|               |          |
|---------------|----------|
| California    | \$7,000  |
| New York      | \$17,600 |
| Texas         | \$9,000  |
| Washington    | \$78,200 |
| Alaska        | \$54,200 |
| Massachusetts | \$15,000 |
| New Jersey    | \$44,800 |
| Pennsylvania  | \$10,000 |

## Tips & Best Practices

- Always upload YTD data before generating results to avoid over-calculating SS tax
- Use Gross Up mode when the employee needs to receive an exact net amount
- Enable Split CSV for large batches to stay under portal upload limits
- Enable Supplemental Rates to auto-fill state withholding percentages
- Check the Dashboard's withholding adjustment explainer when using Custom Data
- Use the client PDF report for stakeholder communication (no tax codes shown)
- Review per-employee tax rates in Settings before generating - auto-populated rates may need adjustment
- The State Tax Export CSV can be fed directly into the State Amendments Automator